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9203/2

Paper 2

Mark scheme

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2 2 B Y 9 2 0 3 / 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	C		1	AO1 3.3.4a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	90°		1	AO1 3.3.4b 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	1°		1	AO3 3.3.4a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	180°		1	AO3 3.3.4a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	(within experimental error), the angle of incidence is equal to the angle of reflection		1	AO3 3.3.4a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	use a greater range of angles of incidence		1	AO4 3.3.4a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	the image in the mirror is laterally inverted		1	AO3 3.3.4c 1–3

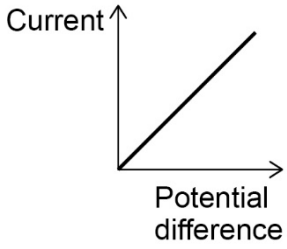
Total Question 1		7
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	voltmeter in parallel ammeter in series	allow 1 mark if the correct components are used the wrong way round	1 1	AO4 3.5.1g 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	12 (V)	allow 12.0 (V)	1	AO2 3.5.1g 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	$R = \frac{1.50}{0.075}$ $R = 20\ (\Omega)$		1 1	AO2 3.5.1h 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	Current  Potential difference		1	AO1 3.5.1i 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	it increases the lifetime of the cell		1	AO4 3.5.1g 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	B		1	AO2 3.5.1r,s 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	C		1	AO2 3.5.1r,s 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	0.75 (V)	this order only	1	AO2 3.5.1r,s 2 × 4–5 1 × 6–7
	0.50 (V)		1	
	1.50 (V)		1	

Total Question 2		12
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	38 56 38	this order only allow 1 mark if 2 are correct	2	AO2 3.7.1d,f 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	high-speed electrons from the nucleus		1	AO1 3.7.2e 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	the proton number increases by one		1	AO2 3.7.2e 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	$\frac{1}{16} = \left(\frac{1}{2}\right)^4$	allow $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$	1	AO2 3.7.2h 6–7
	4 half lives		1	
	$(4 \times 75) = 300$ (s)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	U-235 nucleus absorbs a neutron	do not accept reference to collision	1	AO1 3.7.3c 4–5
	splits into a nucleus of xenon-140 and a nucleus of strontium-94	splits into two smaller nuclei	1	
	releasing (two) further neutrons (and energy)		1	
Total Question 3			10	

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	oscillations / vibrations that are parallel		1	AO1 3.3.1c 4–5
	to the direction of transfer of energy		1	6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	transmitted wave is reflected		1	2 × AO1
	time is measured and multiplied by the speed		1	1 × AO2 3.3.3f,g 1 × 4–5
	time is halved		1	2 × 6–7
	or distance travelled is halved			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	$t = 5.0 \times 10^{-5} \text{ s}$		1	AO3 3.3.3a
	$T = 8 \times 5.0 \times 10^{-5}$	allow 8 x their incorrectly/not converted value of t	1	3.3.3g 2 × 4–5
	$f = \frac{1}{(8 \times 5.0 \times 10^{-5})}$	allow correct substitution of incorrect number of squares and / or and incorrectly / not converted value for t	1	2 × 6–7
		do not allow $\frac{1}{(5.0 \times 10^{-5})}$		
	2500 Hz and no because less than 20 000 Hz	must see 20 000 Hz for this mark to be awarded allow answer consistent with their calculated f	1	

Total Question 4		9
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	higher voltage means lower current (for a given power)		1	AO1 3.6.2b 2 × 4–5 1 × 6–7
	therefore less heating	allow less energy wasted allow reduce power loss	1	
	so it is more efficient		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	$7\,200\,000 = I \times 25\,000$	ignore unit conversion errors	1	AO2 3.6.5c 6–7
	$I = \frac{7\,200\,000}{25\,000}$	ignore unit conversion errors	1	
	$I = 288 \text{ (A)}$	allow 290	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	$I_p \times 400\,000 = 48 \times 25\,000$	ignore unit conversion errors	1	AO2 3.6.2f 6–7
	$I_p = \frac{48 \times 25\,000}{400\,000}$	ignore unit conversion errors	1	
	$I_p = 3 \text{ (A)}$	allow 3.0	1	

Total Question 5		9
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Question 6

Question	Answers	Mark	AO/ Spec. Ref.
06.1	Level 3: The plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced	5–6	AO4 3.1.1h 4–5
	Level 2: The plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content Diagram • clamp, boss and stand • metre rule • band hanging from clamp • top-pan balance • slotted / known masses or newtonmeter. Method • ensure the band is flat • measure the original length of the band with a metre rule • add a known mass • add a slotted mass of 10 g / 100 g / 1 kg • use a top pan balance to measure mass to be added • calculate the weight / force applied to the band • use newtonmeter to measure the force applied • measure new length of band • to calculate extension subtract original length from new length • repeat for more masses • plot graph of force against extension • if directly proportional, graph would be a straight line through the origin.		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	the values in the table are given to 0.01 mm		1	AO4 3.1.1h 1 × 4–5 1 × 6–7
	but a metre rule only measures to 1 mm		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	$X = (2.95 \times 3) - 2.94 - 2.94$		1	AO2 3.1.1h 4–5
	or $X = \frac{(2.94 + X + 2.94)}{3} = 2.95$ $= 2.97$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	<u>x-axis</u> (mean) extension in mm		1	AO3 3.1.1h 3 × 4–5 2 × 6–7 mark with 06.5
	and			
	<u>y-axis</u> force in newtons			
	<u>scale</u> 1.0 mm per major grid line		1	
	<u>plotted points</u> all points plotted correctly	allow 1 mark for 3 points plotted correctly	2	
	<u>line of best fit</u> correct line of best fit drawn with a ruler		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	conversion Δx to metres		1	AO3 3.1.1h $2 \times 4-5$ $1 \times 6-7$ mark with 06.4
	2 pairs of values correctly substituted	allow use of incorrectly/ not converted values of extension	1	
	$k = 140\,000$ (N/m)	allow an answer in the range 130 000 to 150 000 allow an answer consistent with their substituted values	1	

Total Question 6		18
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	$F = 1.8 \times 3.7$	allow 6.7 N	1	AO2 3.1.1e 4–5
	6.66 (N)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	$8.1 = \frac{1}{2} \times 1.8 \times v^2$		1	AO2 3.2.1e 6–7
	$v = \sqrt{\frac{8.1 \times 2}{1.8}}$		1	
	or $v^2 = \left(\frac{8.1 \times 2}{1.8}\right)$			
	$v = 3 \text{ (m/s)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	the moving helicopter has kinetic energy		1	AO3 3.2.2a 3.2.1d,e 3.6.5a 2 × 6–7 2 × 8–9
	the helicopter with a small mass can travel faster for the same kinetic energy		1	
	if the power output from the battery is constant		1	
	the helicopter can travel a greater distance in the same time		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	the centripetal force is less	allow the gravitational force is less	1	AO1 3.8.2f 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	distance from sun is (approximately) 4 times greater		1	AO3 3.1.2c 3.8.2f 8–9
	so the circumference of Mars's orbit is (approximately) 4 times greater than Mercury's		1	
	(the orbital velocity of Mercury is approximately twice the orbital velocity of Mars) therefore $2 \times 4 = 8$	allow 4 times the distance travelled at half the speed takes 8 times longer calculations showing why $T_{\text{mars}} \approx 8 \times T_{\text{mercury}}$ can gain full marks	1	
Total Question 7			13	

Question 8

Question	Answers	Mark	AO/ Spec. Ref.
08.1	Level 3: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5–6	2 × 4–5 4 × 6–7 4 × AO1 3.6.3d, e, f 2 × AO2 3.6.3d, e, f
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3–4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	
	No relevant content	0	
	Indicative content Earth wire - the case is metal - the kettle is not double insulated - so needs an earth wire - if the metal body of the kettle becomes live through a fault - the current is harmlessly conducted away to earth - prevents electrocution if metal case is touched. Fuse - if an electrical fault causes too great a current - the fuse wire will overheat and melt - the live wire is disconnected - which stops the current - prevents overheating / electrocution. Circuit breaker - if an electrical fault causes too great a current - the circuit breaker will switch off (trip) - acts much faster than a fuse - prevents overheating / electrocution.		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	$0.125 = P \times \left(\frac{3.0}{60}\right)$		1	AO2 3.6.5 c, f 8–9
	$P = \frac{0.125}{0.05}$		1	
	(P = 2.5 kW)			
	P = 2500 W		1	
	$2500 = I \times 230$		1	
	$\left(I = \frac{2500}{230}\right)$			
	I = 10.869565		1	
	I = 10.9 (A)		1	
Total Question 8			12	